

Syllabus for M4 Mid-Sem

Logic

1. Propositional Logic
2. Predicate Logic
3. Rule of Inference
4. Boolean Algebra and Circuits
5. Circuit Minimisation - K maps

Proving Techniques

1. Direct Proof
2. Proof by Contradiction
3. Proof by Cases
4. Proof by Contrapositive
5. Proof Strategies

Sets

1. Basic Set Operations - Union, Intersection, Difference
2. Summations
3. Functions - Definition, one-one , onto, bijection
4. Cardinality of sets - Cantor's diagonalization argument, Countable and Uncountable sets, Computable functions and uncomputable functions